# Static versus dynamic estimation of maximal lateral pinch force: implications for design-oriented musculoskeletal modelling

**Joaquín L. Sancho-Bru[1], José L. Iserte-Vilar[1], Néstor J. Jarque-Bou[1], Raquel Lázaro[1]**

[1] *Mechanical Engineering and Construction Department, Universitat Jaume I, Spain, sancho@uji.es*

## Introduction

Musculoskeletal estimates of hand force capacity have traditionally relied on static or quasi-static formulations based on prescribed joint postures [1,2]. Although computationally efficient, these approaches neglect postural evolution during maximal force generation. Forward-dynamic optimization allows muscle activations and posture to evolve [3], but at substantially greater computational cost. This study compared both approaches for maximal lateral pinch force using the same hand–wrist model.

## Methods

The open-access hand–wrist model developed by McFarland et al. [3] was used as the common basis for both formulations. Simulations and optimizations were conducted in MATLAB R2025b using the OpenSim 4.3 API, preserving the same model and parameters except for differences required by each formulation.

In the forward-dynamic formulation, thumb-tip contact was represented by the original kinematic constraint and 0.20 s forward simulations were performed in OpenSim. Fourteen muscle activations were optimized using simulated annealing to maximize mean normal pinch force, penalizing transverse forces exceeding 17% of the normal force, following the original formulation, and wrist excursions beyond ±5°. Fifty independent optimizations were performed ; all met the specified stopping criterion (function tolerance: 0.1; stall limit: 75 iterations).

In the quasi-static formulation, the thumb-tip constraint was replaced by an external force at the interaction point. At each posture, muscle moment arms and active force capacities were obtained from OpenSim, non-muscle generalized forces were estimated by inverse dynamics with muscles disabled, and the generalized effect of the external force was numerically determined. These quantities defined a linear programming problem subject to equilibrium constraints, solved in MATLAB to maximize normal pinch force. Calculations were performed at the common initial posture and at the 50 final dynamic postures. Force estimates and computation times were compared.

## Results

Dynamic optimization produced a broad distribution of maximal pinch forces (57.63 ± 9.36 N), whereas the quasi-static estimate at the common initial posture was 54.46 N. Quasi-static re-evaluation of the 50 dynamically reached final postures yielded 55.89 ± 6.53 N. Paired differences (quasi-static − dynamic) were −1.74 ± 7.05 N (95% CI: −3.75 to 0.26 N), with quasi-static estimates lower in 35/50 runs (Fig. 1). For the five best-performing dynamic solutions, force was 71.70 ± 1.88 N compared with 63.16 ± 1.69 N when the corresponding final postures were evaluated quasi-statically. Median computation time was 2119.79 s per dynamic optimization versus 0.193 s per quasi-static posture evaluation, an approximately 11,000-fold difference under the tested computational conditions.

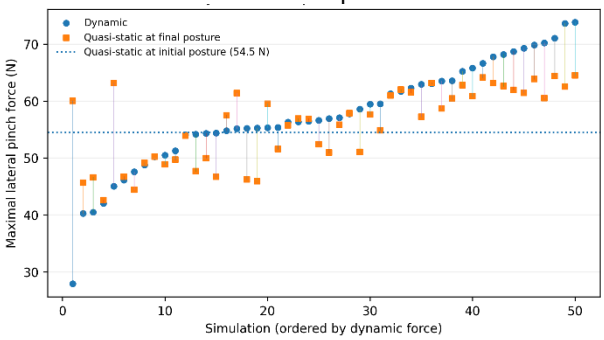



*Figure 1: Paired dynamic and quasi-static estimates of maximal lateral pinch force for the 50 simulations, ordered by dynamic force.*

## Conclusions

Quasi-static and forward-dynamic formulations provided different estimates of maximal lateral pinch force. Postural adaptation did not fully account for these differences, as the highest dynamic forces remained above quasi-static estimates at the corresponding final postures. Residual discrepancies may reflect time-dependent dynamics, differences in penalty implementation, or optimization characteristics, although their relative contributions remain unresolved. Quasi-static analysis nevertheless reduced computation time by four orders of magnitude, making it attractive for design-space exploration where computational efficiency is critical. Whether differences decrease in tasks with more constrained postural freedom, such as power grip, warrants further investigation.

## Acknowledgements

Work supported by Project PID2023-148100OB-I00 funded by MICIU/AEI/10.13039/501100011033 and by FEDER, UE, and by grant number GACUJIMB/2024/08.

## References

1. Wohlman et al, J Biomech, 46: 1014–1020, 2013.
2. Mirakhorlo et al, CMBBE, 21: 548–557, 2018.
3. McFarland et al, IEEE Trans Biomed Eng, 70: 1424–1435, 2023